

MONCHO UNSCHOOLING

Airplanes

Unit Study

“What if the same invisible air that tickles your hand could hold up a machine heavier than a hundred elephants?”

Physics | Math | Language Arts | History | Art & Creativity | Geography | Biology | Technology | Life Skills & SEL | Movement & Body

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🧭 **How to use this guide:** This is not a workbook, and there's no answer key — on purpose! Every challenge is an invitation. Pick the ones that spark curiosity, skip the rest, and go find the answers by doing. **AI plans. Kids DO.** 🚀

✈️ PHYSICS

Airplanes stay up because of invisible air, and with a few sheets of paper you can feel exactly how it works.

🌬️ Challenge 1: Make Paper Lift Off

🎯 Lift and airflow

How can blowing over the top of a paper strip make it rise instead of blow away? ❓

1. Cut a strip of paper about 5 cm wide and 20 cm long. Hold one short end just under your bottom lip, so the strip hangs down and curls away from your chin.
2. Blow a steady stream of air straight across the top of the strip. Watch which way it moves.
3. Fold a second strip into a curved wing: bend it so the top is rounded like a hill and the bottom is flat. Tape the back edges together.
4. Hold your curved wing by the front edge and blow across the top. Try it again with a flat, uncurved strip. Which one lifts higher?
5. Tilt each wing a little more, then a little less, and blow again. Draw a quick table with one row for each wing and tilt, and mark what you saw.

Why: Real airplane wings are curved on top, which is called an airfoil. Fast-moving air over the top has lower pressure (Bernoulli's principle), and the tilt of the wing also pushes air downward, so both help lift the wing up.

🌱 **Make it Easier:** Skip the folded wings. Just blow across one flat strip and describe what happens to the paper.

🚀 **Level it Up:** Make three wings with different curves (a little, medium, a lot). Blow the same way for 5 seconds each and measure the lift in centimeters with a ruler.

🎈 Challenge 2: Balloon Rocket Zipline

🎯 Thrust and force

What makes a balloon dash forward when the air rushes out the back? ❓

1. Thread a long piece of string (about 3 m) through a drinking straw. Tie each end to a chair or doorknob so the string is pulled tight and level.
2. Blow up a long balloon, but do not tie it. Pinch the opening shut and tape the balloon to the straw so it points along the string.
3. Slide the balloon to one end, let go, and watch it zoom. Mark how far it travels with a piece of tape.
4. Clip 1 paper clip to the balloon and launch again. Add a second, then a third. Measure the distance each time.
5. Which load made it go the farthest, and which made it stop soonest? Write down your results.

Why: A jet engine pulls in air, squeezes it, burns fuel with it and blasts hot gas out the back. That gas pushes backward, and Newton's third law says the plane gets pushed forward, just like your balloon.

🌱 **Make it Easier:** Skip the paper clips. Launch the balloon three times and see if it always goes the same distance.

🚀 **Level it Up:** Use a bigger balloon and a smaller one. Predict which will fly farther with 2 paper clips, then test your prediction.

🚗 Challenge 3: Feel the Drag

🎯 Four forces

Why does your hand feel pushed back by air you can't even see? ❓

1. Cut a small card (about the size of your palm). Turn a fan on high and hold the card in front of it. You can also try your flat hand.
2. Hold your hand or the card flat, like a wing, so the air slices over it. Notice how it feels.
3. Slowly turn it upright, like a wall, so the air hits the whole face. How does the push change?
4. Tilt it a little upward. Do you feel it want to rise? That is lift. Feel the push backward too? That is drag.
5. Draw a plane and label four arrows: lift, weight, thrust and drag. Make each arrow longer or shorter based on what you felt.

Why: Every airplane flies by balancing four forces: lift, weight, thrust and drag. The more surface faces the air, the more drag you feel.

🌱 **Make it Easier:** Only try two angles, flat and upright, and tell a grown-up which one pushed back harder.

🚀 **Level it Up:** Cut the card into three sizes (small, medium, large) and test each at the same angle to find out how size changes drag.

➗ MATH

Numbers help us measure how far, how fast, and how to point a plane the right way.

➗ Challenge 1: Measure the First Flight

🎯 Measuring Length

How many of your own steps would it take to walk the whole first airplane flight? ❓

You're the surveyor at Kitty Hawk, North Carolina, and you must mark the path of the very first powered flight on December 17, 1903. It covered about 37 m and lasted about 12 seconds.

1. Use a tape measure to mark 1 m on the floor. Walk it and count your steps. Do it three times and use your usual number.
2. Multiply your steps per meter by 37 to guess how many steps the flight covers.
3. Stick a strip of tape at the start. Walk your steps and place tape at the end.
4. A Boeing 747 jumbo jet has wings about 64 m across. Use your step count to measure 64 m. Would those wings even fit along the whole first flight?

🌱 **Make it Easier:** Measure with your feet heel-to-toe instead of steps. Round your count to the nearest 5.

🚀 **Level it Up:** Time how long you need to walk the path. Compare it to the 12 seconds the flight took. Who was faster, you or the airplane?

➗ Challenge 2: Race Across the Atlantic

🎯 Speed, Distance and Time

How much of that long, lonely flight could a modern jet save? ❓

In May 1927, Charles Lindbergh flew alone from New York to Paris, about 5,800 km in about 33.5 hours. To find speed, divide distance by time. To find time, divide distance by speed.

1. Divide 5,800 by 33.5 to find Lindbergh's average speed. Round to the nearest whole number.
2. A jet flies about 900 km per hour. Divide 5,800 by 900 to find the time.
3. Subtract the jet's time from 33.5 hours. How many hours were saved?
4. Write your answers in a neat list with a unit next to every number.

Why: Speed means how much distance you cover in one hour, so the same distance takes less time when the speed goes up.

🌱 **Make it Easier:** Round Lindbergh's flight to 30 hours and the jet's speed to 1,000 km per hour.

🚀 **Level it Up:** Turn the jet's decimal time into hours and minutes. Then try a flight from your own city to a place of your choice, using the same jet speed.

➗ Challenge 3: Number a Runway

🎯 Scaling Numbers

Why would a runway have a number that comes from a compass? ❓

Pilots and air traffic controllers use runway numbers to know which way to line up. A runway's number comes from its compass heading: take the degrees and drop the last zero. Runway 27 points toward 270 degrees, which is roughly west.

1. Draw a big circle on paper and mark N at the top (0 degrees), E on the right (90), S at the bottom (180) and W on the left (270).
2. A plane heading east flies at 90 degrees. Drop the zero to find its runway number.
3. Find the numbers for south (180) and for north (360). Check the pattern.
4. Draw a runway as a long rectangle and write its number at each end. The opposite end of a runway differs by 180 degrees, so add or subtract 18 from the number.

🌱 **Make it Easier:** Only number the four main directions: north, east, south and west.

🚀 **Level it Up:** Add a second runway pointing northeast (45 degrees). Work out its two numbers. Round to the nearest whole number if needed.

📚 LANGUAGE ARTS

Words are the tools pilots, controllers and word-lovers use to talk about the sky.

🎙️ Challenge 1: Control Tower Calling!

🎯 Phonetic Alphabet

How could you spell a word over a noisy radio without anyone mixing up "B," "D" and "P"? ❓

You're the control tower operator, and your partner is the pilot. Pilots and air traffic controllers use the phonetic alphabet (Alpha, Bravo, Charlie...) so letters are not confused over the radio.

1. Ask an adult to poke a hole in the bottom of two paper cups with a pencil, then join them with a long string. Pull it tight to make your pretend radio.
2. Write the alphabet down the page, then pick a word like Alpha for A, Bravo for B and Charlie for C. Make up your own for the rest, or look them up in a book.
3. Spell your name to your partner through the cups, using only the code words.
4. Pick three secret words and spell each one out. Your partner writes down the letters and reads the words back.
5. Swap jobs and do it again.

Why: Many letters sound alike, like B, D and P. A whole word is much harder to mishear than one letter.

🌱 **Make it Easier:** Use only the first 10 letters of the alphabet, and spell short three-letter words like "cat."

🚀 **Level it Up:** Make a chain of three partners. The first one spells a secret message to the second, who passes it on to the third. Compare the last message with the first one.

🔤 Challenge 2: Word Detectives in the Sky

🎯 Word Roots

What did people call a flying machine before anyone knew how to build one that worked? ❓

The word "aeroplane" is often explained as Greek "aero" (air) plus "plane," from Latin "planum" (flat surface). "Aviation" comes from Latin "avis," meaning bird.

1. Fold a sheet of paper into a little book with 4 pages.
2. On page 1, write "aero = air" and "planum = flat surface." Under them, write "avis = bird."
3. Explain in your own words why "air + flat surface" is a good name for a plane. Why might "aviation" mean "bird-like"?
4. Invent three new flying-related words by mixing the roots, like "aerohouse" or "avisport." Write each one with its root meanings and what it could be.
5. Draw a tiny picture next to each new word.

🌱 **Make it Easier:** Invent just one new word and draw it.

🚀 **Level it Up:** Use a dictionary to find two real English words that use "aero" or "avi," like "aerobic" or "aviary," and add them to your mini-dictionary with their meanings.

📓 Challenge 3: Alone Above the Ocean

🎯 Diary Writing

What would your mind do after 33 hours with no one to talk to and no sleep? ❓

You're Lindbergh's journal-keeper, writing from the cockpit of his lonely flight across the Atlantic. Lindbergh flew alone for about 33 hours and could not sleep.

1. Grab paper and a pencil and write "Dear Diary" at the top.
2. Write in first person ("I"). Use real details: 33 hours alone, no sleep, the cold, the noise of the engine and the endless ocean below.
3. Describe at least two things you see, one thing you hear and one feeling you have.
4. Include one moment when it gets very hard to stay awake. What do you do?
5. End with what you hope to see when the flight is finally over.

🌱 **Make it Easier:** Write just five sentences, each starting with "I see," "I hear," "I feel," "I want" and "I hope."

🚀 **Level it Up:** Write a second entry from later in the flight, when you are more tired, and show the change in your thoughts through your word choices.

📜 HISTORY

Long before jets crossed the sky, people in history figured out how to fly, one daring idea at a time.

🌬️ Challenge 1: Build a Wind Tunnel Like the Wright Brothers

🎯 Testing Ideas

How could two bicycle makers learn what a wing needs without ever leaving the ground? ❓

You're the Wright brothers' mechanic in 1901, and the brothers have asked you to build a wind tunnel. They used a wooden box with a fan and small metal wing models inside it to test wing shapes.

1. Cut a window in the side of a long cardboard box and cover it with clear plastic wrap so you can see inside. (Ask a grown-up to help with cutting.)
2. Set a small fan at one open end of the box.
3. Cut three paper wings: a flat strip, a gently curved strip, and a very curved strip. Tape each one to a straw so it can stick out in the airflow.
4. Hold each wing in the tunnel and watch it. Does it rise, wobble or flop? Give each shape a score from 1 to 5.
5. Tape your tunnel together so it can be used again.

Why: Air pushes differently on different shapes, so testing small models showed the brothers which wings worked before they built full-size ones.

🌱 **Make it Easier:** Test just two wings, a flat one and a curved one, and hold them in front of the fan without building the box.

🚀 **Level it Up:** Test each wing at three different angles, tilted up a little, a lot, or not at all, and make a chart of your scores.

🎤 Challenge 2: Interview Bessie Coleman

🎯 Asking Questions

What would you do if no flight school in your country would let you in?❓

Bessie Coleman was born in Texas in 1892. In 1921 she became the first African American woman to earn a pilot's license, and she earned it in France.

1. Read a short biography of Bessie from a library book or a kids' encyclopedia. Write down 3 things you learn.
2. Write 5 interview questions, such as why she went to France, what she had to learn and how she paid for it.
3. Decide who plays the reporter and who plays Bessie. Your partner can use your notes to answer in her voice.
4. Act out the interview, then switch roles and do it again.
5. Add one question you wish you could ask her.

🌱 **Make it Easier:** Ask only 3 questions and let your partner answer from a short list of facts you write together.

🚀 **Level it Up:** Invent a second interview with a person who helped Bessie, and decide what they might say about her.

⚖️ Challenge 3: First in the Sky? The Great Flight Debate

🎯 Making an Argument

What counts as the first flight, floating in a balloon or flying in an airplane? ❓

In 1783 the Montgolfier brothers launched the first crewed hot air balloon flight in France, long before airplanes existed. Leonardo da Vinci had drawn flying machines hundreds of years earlier, and the Wright Flyer took off in 1903.

1. Make three teams, or take turns if you have fewer players. Each team picks one: the 1783 balloons, da Vinci's sketches or the 1903 Flyer.
2. Look up one simple fact about your choice in a book or encyclopedia.
3. Give each team 2 minutes to argue why its choice came first and mattered most.
4. Let each team ask one question of another team.
5. Vote with your eyes closed. You can't vote for your own team.

🌱 **Make it Easier:** Debate just two choices, balloons against the Flyer, and give each side one reason.

🚀 **Level it Up:** Add a fourth choice, such as kites or gliders, and invite someone to be the judge who must explain the winner.

🎨 ART & CREATIVITY

Art lets you imagine, design, and paint the invisible air and the machines that ride on it.

🪶 Challenge 1: The Apprentice's Flying Machine

🎯 Observational Drawing

How could a bird's bending wing become the blueprint for a machine you invent? ❓

You're an apprentice in Leonardo's workshop, and today the master wants new ideas. Around 1485-1505, Leonardo da Vinci filled notebooks with flying machines, including an ornithopter that flapped its wings, inspired by bird flight.

1. Look at a bird out your window, or find a photo of one in a book. Notice how its wing bends when it flaps.
2. Sketch your own flying machine with a pencil on plain paper. Copy the curve of that bird wing.
3. Add small labels with arrows to show which part moves and which part holds still.
4. Write your notes in mirror-writing, from right to left, so they only read correctly when you hold the page up to a mirror.

Why: Leonardo studied birds because their wings change shape in the air, and he hoped a machine could copy that trick.

🌱 **Make it Easier:** Skip the mirror-writing and label your drawing in normal writing, with just two labels.

🚀 **Level it Up:** Draw a second view of your machine from above, and write a note explaining what you would change if a test flight failed.

✈️ Challenge 2: Airline Livery Designer

🎯 Design Planning

Can a plane look amazing and still fly straight? ❓

Real airplanes wear a livery, a paint scheme with colors and logos that makes each airline easy to spot. Paper airplanes can fly in contests too, and the Guinness record throw is over 69 m, set with a specially folded plane. Your job is to design a plane that looks great and flies.

1. Invent an imaginary airline. Pick a name, two or three colors, and a simple logo.
2. Fold a sheet of paper into a paper airplane and make a test throw before decorating.
3. Unfold it and sketch your livery lightly in pencil. Keep the design flat and even on both wings.
4. Color it with markers or crayons, refold it along the same creases, and test again. If it flops, adjust the folds.

Why: Heavy blobs of paint or tape on one wing make the plane lean to that side, so spread your decorations evenly.

🌱 **Make it Easier:** Decorate with just one color and one simple logo on the tail.

🚀 **Level it Up:** Make two planes with different liveries, and see which one flies farther over three throws each.

☁️ Challenge 3: The World From 10,000 Meters

🎯 Layering Colors

What colors would you see if you looked down from a window seat above the clouds? ❓

A typical passenger jet cruises at about 900 km/h at roughly 10,000-12,000 m, about 33,000-39,000 ft. From a window seat, the ground looks like a patchwork far below, with clouds drifting in between.

1. Paint the bottom layer first with watercolors or diluted paint: green and brown patches for fields, blue for a lake or sea.
2. Let it dry for a few minutes.
3. Dab a sponge in white paint and press it on the page to make fluffy clouds. Overlap some clouds so the land peeks through the gaps.
4. Paint a deep blue strip across the top of the page for the sky above you, fading lighter toward the clouds.

Why: Things far away look smaller and paler, so painting distant parts lighter makes your picture feel deep.

🌱 **Make it Easier:** Use only three colors: green, blue, and white. Dab just a few clouds.

🚀 **Level it Up:** Add a tiny wing of the plane at the edge of your picture, and use lighter shades for the land that is farthest away.

🌍 GEOGRAPHY

Every airplane trip is a geography puzzle, because pilots must find the best path across a round planet.

🧵 Challenge 1: The String Route

🎯 Map Reading

Why would a plane flying from New York to Tokyo fly toward Alaska instead of straight across the middle of the map? ❓

You're the route planner for a flight from New York to Tokyo. Here's your job:

1. Find New York and Tokyo on a globe and stick a small piece of tape on each one.
2. Pull a piece of string tight between the two tapes so it hugs the globe. That is a "great circle" route, the shortest path over a round Earth.
3. Notice which places the string passes near, then pick it up without moving the tapes.
4. On a flat world map, draw a straight line between the same two cities.
5. Lay your string on the flat map. Does it match your straight line?

Why: A globe is round, but a flat map stretches it out, so the shortest path looks curved on paper.

🌱 **Make it Easier:** Pick two cities that are closer together, like New York and London, and just look at where the string goes.

🚀 **Level it Up:** Make a second string route from Los Angeles to London, then guess which route is longer before you measure both strings against a ruler.

🛫 Challenge 2: Airport Code Match-Up

🎯 Place Knowledge

How can just three letters tell a pilot exactly which airport in the world to land at?❓

Airports use three-letter codes. JFK is New York's John F. Kennedy Airport, LHR is London Heathrow, and NRT is Tokyo Narita. Time to play with them:

1. Get a blank world map (or sketch a simple one) and write JFK, LHR and NRT on small paper slips.
2. Place each slip on the correct city. Use an atlas or globe to check yourself.
3. Look up three more codes with a grown-up, such as LAX, CDG and SYD, and match those too.
4. Invent a code for your own town by choosing three letters that sound or look like its name.
5. Mix up all the slips and challenge a friend or family member to match them.

🌱 **Make it Easier:** Match only the three codes from this challenge and use a map with the city names already written on it.

🚀 **Level it Up:** Invent codes for five places you love, then see whether anyone can guess the town from the letters alone.

🗺️ Challenge 3: Round-the-World Dream Trip

🎯 Trip Planning

If you fly across the whole world, how many times will your watch be wrong about what time it is?❓

Design a dream trip around the world using a round-the-world ticket:

1. Draw a big world map on paper or use a printed one.
2. Choose five airports, one on each of five different continents. Mark each with a dot and write its name beside it.
3. Connect the dots in the order you will fly, with an arrow for each flight.
4. Guess the time zones you'll cross. When you leave home at noon, what time do you think it is at each stop? Write your guess next to every dot.
5. Check your guesses with a time zone map or a grown-up's phone clock, and fix any that were off.

🌱 **Make it Easier:** Choose only three airports on three continents and check just one time zone.

🚀 **Level it Up:** Add a stop in the Southern Hemisphere, plan which season you'd find at each airport, and count how many days your whole trip would take.

🌿 BIOLOGY

Long before engineers drew a single wing, living things were already mastering the air, and their designs still teach us how flying works.

🪶 Challenge 1: Feather vs. Paper Wing

🎯 Observation

Why can a feather float so gently when it weighs almost nothing? ❓

Take a real feather (or look closely at a photo of one) outside to a calm spot.

1. Hold the feather up to the light. Notice how it curves, how thin and light it is, and look for the tiny hairs that zip together along its edges.
2. Fold a strip of paper the same length so it bends in a gentle curve like the feather.
3. Stand on the same spot and release the feather and the curved paper from the same height. Do it three times and watch which glides farther.
4. Sketch both side by side and label what you notice: shape, lightness, curve.

Why: a curved, light shape gives the air more time to push up on it, so it slows its fall and slides sideways instead of dropping straight down.

🌱 **Make it Easier:** Skip the paper bending. Just drop the feather and a flat piece of paper and draw which one floated longer.

🚀 **Level it Up:** Add a paper clip to each one and test again. How much weight can each carry before it stops gliding?

🌳 Challenge 2: Seed Flyers Hunt

🎯 Classification

How can a tiny seed travel far from its tree without any wings or engine? ❓

Nature invented flying machines long before people did. Maple "helicopter" seeds spin as they fall, so the wind can carry them far from the parent tree. Go outside or to a park and hunt for seeds that fly, spin or float: maple keys, dandelion fluff, pine seeds, milkweed. No seeds nearby? Look at photos in a free plant book or a Wikimedia Commons search.

1. Collect 3-4 different seeds.
2. Stand on a sturdy chair or the stairs, with an adult nearby, and drop each seed from the same height. Watch how it falls: spinning, drifting or plummeting.
3. Cut a paper strip about 3 cm x 12 cm. Fold the bottom up to make a weight, then split the top into two flaps and fold them in opposite directions.

4. Drop your paper spinner, then tweak it: a longer wing, a shorter wing, a paper clip on the bottom.
5. On a field page, draw each seed, note how it fell (spin, drift or drop), and sketch your spinner with the one change that made it spin best.

Why: a spinning wing keeps moving through the air even while it falls, which slows the drop and gives the wind more time to carry it away.

🌱 **Make it Easier:** Test only two seeds and the paper spinner, and just circle "spin," "drift" or "drop" for each one.

🚀 **Level it Up:** Drop each seed three times and count the seconds it takes to land. Which one stays in the air longest, and does your spinner beat it?

💻 TECHNOLOGY

Airplanes are machines you steer with tiny moving flaps, protect with smart design, and guide with precise instructions.

🎛️ Challenge 1: Build the Test Pilot's Control Panel

🎯 Systems and Controls

How can a few little flaps turn a giant plane through the air?❓

You're the test pilot, and your plane needs a control panel before takeoff. Planes use three control surfaces: ailerons make the plane roll, elevators make it pitch up or down, and the rudder makes it yaw left or right.

1. Cut a cardboard rectangle and draw three labeled controls on it: two levers and a pedal.
2. Cut out a paper airplane shape. On each wing's back edge, cut a small flap for the ailerons. Cut flaps on the tail's back edge for the elevators, and one on the upright tail fin for the rudder.
3. Fold each flap with a crease, so it works like a hinge.
4. Hold the plane and move one flap at a time. Blow gently from the front and watch which way the nose or wings want to move.
5. Match each flap to its lever on your control panel.

🌱 **Make it Easier:** Make only two flaps, the elevator and the rudder, and test them one at a time.

🚀 **Level it Up:** Move the left aileron up and the right aileron down at the same time. Predict what the plane will do before you blow, then check.

📦 Challenge 2: Black Box Challenge

🎯 Protective Design

What would your invention need so it survives a crash and can still be found? ❓

Real airplanes carry a flight recorder called the black box, and it is actually painted bright orange so it is easier to find after an accident. Your job is to design a container that protects a secret message.

1. Write a secret message on a small strip of paper.
2. Choose materials to protect it: a small box, tissues, cotton balls, bubble wrap, a sock, tape.
3. Pack the message so it can't rattle around, and color the outside bright orange or another super-bright color.
4. Drop it from the seat of a chair onto the floor. Check whether the box stayed shut and the message is safe.
5. Improve your design and drop it again from a higher spot, like a table, with an adult's help.

Why: Soft padding makes the box slow down more gently when it lands, so less force reaches the message inside.

🌱 **Make it Easier:** Use only one protective material and a single drop from your lap height.

🚀 **Level it Up:** Build two different designs, drop both from the same height, and decide which protects better.

💻 Challenge 3: The Folding Code

🎯 Algorithms

Can you write instructions so clear that someone else folds your plane without ever seeing it? ❓

Programmers write algorithms, which are exact step-by-step instructions a computer can follow without guessing.

1. Fold a paper plane you know well, and pay attention to each fold as you go.
2. Write your folding code, one action per step. Use clear words like "fold the top corners to the center line" and number every step.
3. Hide your plane. Give your code to a friend or family member with a sheet of paper.
4. Watch silently as they follow it, without helping or pointing.
5. Compare their plane to yours. Wherever they got stuck or folded something different, fix that step in your code.

🌱 **Make it Easier:** Write only the first five folds and add small drawings next to each step.

🚀 **Level it Up:** Add symbols, like arrows for folds and dotted lines for creases, so your code works without words.

🧘 LIFE SKILLS & SEL

Flying is as much about calm, clear teamwork as it is about wings, so these challenges train your heart, your voice, and your crew.

🎭 Challenge 1: Flight Attendant on Duty

🎯 Clear Communication

How do you teach a plane full of people something important without making them panic? ❓

You're the flight attendant, and your family just boarded Flight Moncho 101. Airplanes require seat belts and safety demonstrations, so it's your job to give one.

1. Set up rows of chairs or cushions and have your family "buckle in."
2. Plan three moves: how to fasten a seat belt, where the exits are, and how to use the oxygen mask.
3. Practice using a slow, calm, friendly voice and big, clear hand motions.
4. Tell your passengers to put their own oxygen mask on first before helping anyone else, and say it out loud as a rule.
5. Perform your demo, then ask one passenger to repeat the mask rule back to you.

Why: you can't help anyone if you run out of air yourself, so taking care of yourself first keeps you able to help others.

🌱 **Make it Easier:** Practice with just two moves, the seat belt and the oxygen mask, and read them from a short note card.

🚀 **Level it Up:** Add a safety card with simple drawings for each rule, and add a "what if" question from a passenger that you answer calmly.

✍️ Challenge 2: Courage Before Takeoff

🎯 Managing Big Feelings

What can you do with your breath and your words when your tummy is full of butterflies? ❓

Pretend it's your very first flight and your stomach is fluttering. Let's practice being brave.

1. Sit in a "seat" and imagine the engines starting to roar.
2. Do calm breathing: breathe in slowly through your nose for 4 counts, then out through your mouth for 4 counts. Repeat 5 times.
3. Say your worries out loud, like "I'm scared of bumps" or "I'm nervous about the noise," and write each one on a scrap of paper.
4. Next to each worry, write one thing that helps, like squeezing a stuffed animal, counting breaths, or looking out the window.
5. Copy your best helpers onto an index card and decorate it with a cloud or a plane.

🌱 **Make it Easier:** Name just one worry and draw a picture of the thing that helps instead of writing it.

🚀 **Level it Up:** Write a short pep talk to your future self, starting with "When I feel nervous, I will..." and teach your breathing to a family member.

🤝 Challenge 3: Crew Handoff Mission

🎯 Teamwork and Listening

What happens to a team when everyone has a job and everyone waits their turn to talk? ❓

Your crew has one mission: get a paper airplane safely across the room.

1. Gather three or more crew members and one paper airplane. Assign roles: the pilot (holds and launches the plane), the tower (gives the go-ahead), and the navigator (calls out where to aim).
2. Make a rule: only the person whose job it is can speak, and no one talks over anyone else.
3. The navigator calls the target, the tower says "clear for takeoff," and the pilot launches.
4. If someone talks out of turn, the plane goes back to the start.
5. Switch roles and try again until every crew member has done each job.
6. Sit together and share one thing that went well and one thing to change.

🌱 **Make it Easier:** Use only two roles, a pilot and a tower, and let the plane travel a short distance.

🚀 **Level it Up:** Add a hidden target, like a laundry basket, and have the crew plan out loud for one minute before the first launch.

🤸 MOVEMENT & BODY

Your body is a flying machine, and with the right moves you can steer, power and land it just like an airplane.

🤸 Challenge 1: Human Airplane Control Room

🎯 Body Control

Can you turn your own body into a plane that obeys every command? ❓

You and a partner are a flight team. Stand with your arms out wide as wings. Your partner is the control tower and calls the moves.

1. Roll: tilt your arms like a seesaw, one wing up and one wing down, without bending your legs.
2. Pitch: tip your whole body forward (nose down), then lean back a little (nose up).
3. Yaw: keep your arms level and swivel your feet and hips left and right, like a nose pointing to a new direction.
4. Your partner calls a mix of three commands in a row ("roll right, pitch up, yaw left") and you perform them smoothly.
5. Swap jobs, then make up a 6-move routine and run it with no pauses.

🌱 **Make it Easier:** Practice just one move at a time, and let your partner call only one command per turn.

🚀 **Level it Up:** Add a fourth command, like "bank and turn" (roll and yaw together), and try to run your routine with eyes closed while your partner guides you.

🤸 Challenge 2: Flap Like a Bird Endurance Test

🎯 Heart Rate and Muscles

Why do birds have such giant chest muscles, and how would your own body cope with flapping? ❓

Birds have giant chest muscles because flapping is hard work. Find out how your own body handles it.

1. Rest for two minutes. Then find your pulse on your wrist or neck, count beats for 15 seconds and multiply by 4. This is your resting pulse.
2. Flap your arms like a bird for 1 minute without stopping. Right away, count your pulse again for 15 seconds and multiply by 4.
3. Sit and rest. Check your pulse every 30 seconds to see how long it takes to get back to normal.
4. Try other bird moves, resting until your pulse is back to normal between each one: slow flapping, small flapping and flapping while hopping. Repeat the pulse steps for each.

5. Notice which muscles burn: shoulders, upper back, chest. Make an endurance log chart with your resting pulse, post-flap pulse and recovery time for each move, then draw a body outline and color the muscles that got tired.

Why: Your muscles use more oxygen when they work hard, so your heart beats faster to deliver it.

🌱 **Make it Easier:** Test only two bird moves, like regular flapping and slow flapping, and skip the recovery timing.

🚀 **Level it Up:** Predict which move will raise your pulse the most before you test, then see if you were right and find out why.

🤸 Challenge 3: Parachute Drop and Bedsheet Dome

🎯 Coordination

How can a piece of cloth and a team of friends teach your body to move with the air? ❓

A parachute works by catching air to slow a fall. First, build a mini one.

1. Cut a square from a cloth napkin or plastic bag. Tie one string to each corner.
2. Tie the four strings together at the bottom and attach a small toy as the passenger.
3. Drop it from a safe height with an adult, like a step or a sturdy chair, and watch how it falls. Then make the canopy bigger or smaller and drop it again to see how the fall changes. Film or photograph each drop.
4. Now for the big version. Grab a bedsheet and a few friends or family members. Everyone holds an edge and lifts together to fill the sheet with air, making a big dome.
5. Take turns ducking under the dome and running to a new spot before it falls. Notice how everyone has to move in sync. Then invent your own bedsheet parachute game and write a list of its rules.

Why: A bigger canopy catches more air, so more air pushes back against the fall and slows it down.

🌱 **Make it Easier:** Skip the canopy size test and just drop one parachute, then play the bedsheet game with a simple rule: lift, lower, run.

🚀 **Level it Up:** Add a challenge to your game, like having each runner switch places with a partner or carry a soft toy across without letting it touch the ground.